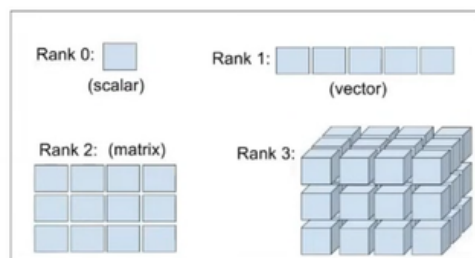




Tensor is specialized multi-dimensional array designed for mathematical and computational efficiency.



Creating a Tensor

```
torch.empty(2, 3)
torch.zeros(2, 3)
torch.ones(2, 3)
torch.rand(2, 3)
torch.manual_seed(100)
torch.rand(2, 3)
torch.tensor([[1, 2, 3],[4, 5, 6]])
torch.arange(0, 10, 2)
torch.linspace(0, 10, 5)
```

Tensor Shapes & Data Types

```
x = torch.tensor([[1, 2, 3],[4, 5, 6]])
x.shape
torch.empty_like(x)
torch.zeros_like(x)
torch.rand_like(x)
torch.rand_like(x, dtype=torch.float32)
x.dtype
torch.tensor([1.0, 2.0, 3.0, 4.0],
              dtype=torch.int32)
```

Scalar Operation

```
x + 2
x - 2
x * 3
x / 4
x // 5
x ** 2
```

Element-wise operation

```
a = torch.rand(2, 3)
b = torch.rand(2, 3)
c = torch.tensor([1, 2, -3, 5, -6])
d = torch.tensor([1.2, 3.8, 4, 5.6])

a + b          torch.abs(c)
a - b          torch.neg(c)
a * b          torch.round(d)
a / b          torch.ceil(d)
a ** b         torch.floor(d)
a % c          torch.clamp(d, min=2, max=3)
```

Reduction Operation

```
e = torch.randint(size=(2, 3), low=0,
                  high=10, dtype=torch.float32)

torch.sum(e)          torch.min(e)
torch.sum(e, dim=0)   torch.max(e)
torch.sum(e, dim=1)   torch.prod(e)
torch.mean(e)         torch.std(e)
torch.median(e)       torch.argmax(e)
torch.min(e)          torch.var(e)
```

<https://github.com/Rudra-G-23/deep-learning-using-pytorch>

